Examine.com

Memory & Focus Stack Guide

Written by the editors of Examine.com

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Medical Disclaimer

This guide is a general health-related information product intended for adults over the age of 18.

This guide is for educational purposes only. It does not constitute medical advice. Please consult a medical or health professional before you begin any exercise, nutrition, or supplementation program, or if you have questions about your health.

If you choose to engage in any activity or use any product mentioned in this guide, you do so of your own free will, and you knowingly and voluntarily accept the risks.

While we mention major known interactions, it is possible for any supplement to interact with other supplements, as well as with foods and pharmaceuticals. Therefore, it is important to consult a medical professional prior to using any supplement mentioned in this guide.

Specific study results mentioned in this guide should not be considered representative of typical results. Individual results do vary.

Keep in mind that not all supplements contain the exact compounds and amounts listed on the label. Always investigate supplement companies, as well as the supplement itself, before purchasing anything. Also note that, more than isolated compounds, herbs can have variability from batch to batch, which can alter their efficacy and safety.

For evidence supporting the claims mentioned in this guide, please visit Examine.com.

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How to Use This Guide

The team at Examine.com has been publishing research on nutrition and supplementation since March 2011. Drawing from all we've learned, we designed this Stack Guide to help you figure out which supplements can help you reach your health goal, and which can hinder you or just waste your money.

Core supplements are the supplements most likely to help, while having little to no side effects. They tend to have more research backing than do the other supplements.

Primary options may provide substantial benefit, but only in the right context. A primary option is not for everyone, but if you read the entry and find that you meet the criteria, consider adding the supplement to your stack.

Secondary options form another group of potentially beneficial supplements, but with less evidence for their effects. They could work or be a waste of money. Keep them in mind, but think twice before incorporating them into your stack.

Inadvisable supplements have either been shown to be ineffective, marketing claims notwithstanding, or are considered too risky. Do not add them to your stack. At best, they will be a waste of money; at worst, they can cause you harm.

Now that you have been presented with various supplements worthy of your interest, the time has come to combine them based on your objective. We will guide you in **assembling your stack**.

Then comes the **FAQ**, in which we cover common questions that may arise when assembling your stack.

Lastly, we include information on **precautions and troubleshooting**.

With all this combined, you should be able to identify and assemble the supplement stack best suited for your objectives.

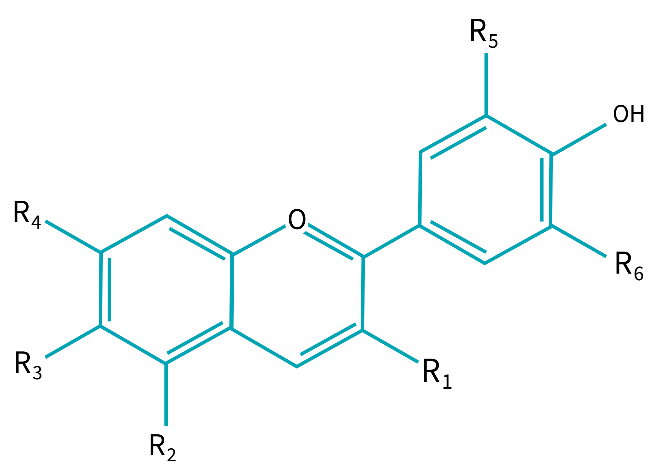
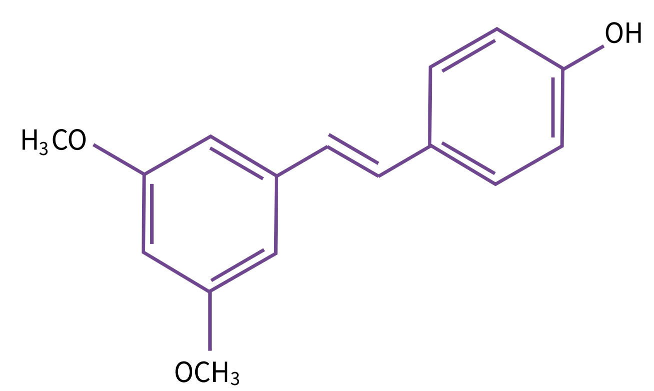
Core Supplements

Blueberry

Why it's a core supplement

Blueberries and other dark berries contain anthocyanins and pterostilbene. These molecules can protect the brain and influence its activity.

Figure 1: The differences between Anthocyanins and Pterostilbene

Anthocyanins	Pterostilbene
	
Pigments that often give a red, purple, or blue color to the plants that have them	A derivative of resveratrol, the antioxidant found in red wine
Some have antioxidant or anti-inflammatory properties	May possess antioxidant and anticancer properties
More than 635 anthocyanins have been identified	Can help reduce blood glucose and improve insulin sensitivity

Blueberries can also increase the activity of a neuropeptide known as neuronal growth factor (NGF), probably due to their anthocyanin content. When neurons grow, they branch out toward each other. NGF helps neurons grow, which makes it easier for them to communicate.

Though the only human evidence for blueberries increasing cognition comes from studies on senior citizens, the mechanism and animal evidence suggest that this fruit can also benefit the brain of healthy people.

Blueberries do not interact negatively with pharmaceuticals. Safe and readily available, they are the ideal core supplement for a *Memory & Focus* stack.

How to take it

Studies support the following protocols:

- Blueberry anthocyanins: 500–1,000 mg/day.
- Blueberry powder: 5.5–11 g/day (≈ 55–110 mg of anthocyanidins).
- Fresh blueberries: 60–120 g/day (≈ 100–200 mg of anthocyanidins).
- Blueberry juice: 500 mL/day. Blueberries should be the first ingredient of the juice. Any juice made primarily from sugars, with added flavoring, will not provide the same benefits to memory or focus.

Blueberries can be swapped for other dark berries, such as blackberries or bilberries. Just remember that most studies are on blueberries and that the amount of anthocyanins (or of anthocyanidins, which are the anthocyanins without their sugar molecule) does not tell the whole story, if only because different *kinds* of anthocyanins can be found in different amounts in different berries. Most importantly, dark berries and red berries have very different anthocyanin profiles.

Primary Options

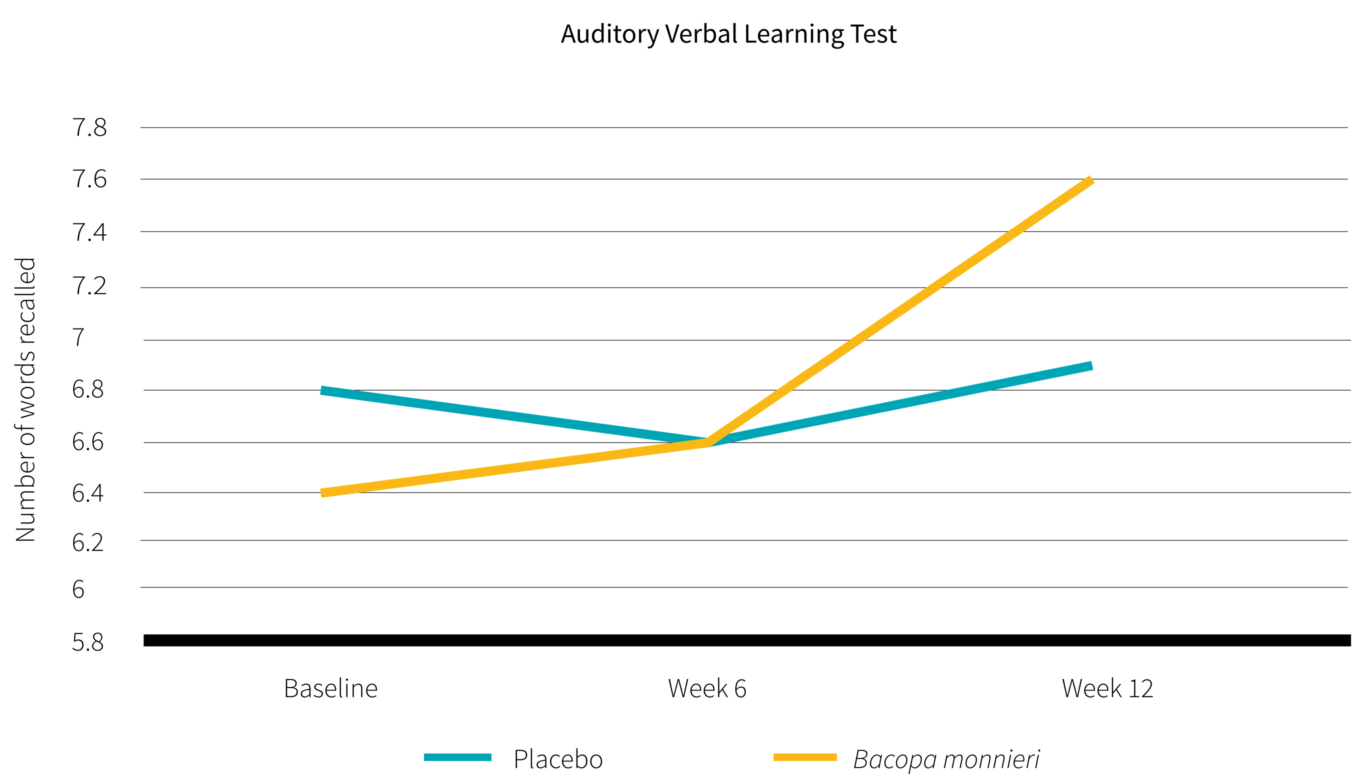
Bacopa Monnieri

Why it's a primary option

Bacopa monnieri is a swamp plant used in traditional Indian medicine to improve memory and cognition. While most effective for elderly people, it can provide benefits for all age groups, but only after one month of supplementation.

Figure 2: Bacopa monnieri's memory-enhancing effects

Extended supplementation with *Bacopa monnieri* may be able to enhance memory recall. During an Auditory Verbal Learning Test (AVLT), the group taking Bacopa monnieri was able to recall more words read off to them from a list of 15 words.



Source: Calabrese et al. *J Altern Complement Med*. 2008 Jul.

Bacopa monnieri reliably improves working memory (your RAM, so to speak, which determines how much information you can keep at the forefront of your mind). Further research is needed to determine if it can affect verbal fluency, word processing, and attention span.

Bacopa monnieri is not a core supplement because more research is needed to ascertain that it does not interact negatively with pharmaceuticals.

How to take it

Take a dose once a day with food. Take 150 mg of bacosides, the active compound in *Bacopa monnieri*, so about 300 mg of an extract with a 55% bacoside content. To supplement the leaf powder, assuming a 10–20% bacoside content, take 750–1,500 mg. Further research is needed to determine the efficacy and safety of higher doses.

Caffeine with Theanine

Why it's a primary option

Theanine is an amino acid with relaxing, but not sedating, properties. It neither interacts with sedative neurotransmitters nor causes feelings of fatigue. It can reduce the overexcitability often caused by caffeine without impairing caffeine's stimulatory effect. In fact, the improvements in concentration (focus and attention span) induced by caffeine on the one hand and theanine on the other have been shown to be synergistic.

Even though coffee is a popular beverage worldwide, caffeine is not innocuous. Regular consumption leads to tolerance and often to dependence and withdrawal. Moreover, caffeine interacts dangerously with monoamine oxidase inhibitors (MAOIs), a kind of antidepressant, and several other pharmaceuticals, notably tizanidine.

Caffeine can also decrease blood lithium levels. Suddenly eliminating all caffeine from your diet can cause lithium levels to rise. If you are on lithium, keep your day-to-day caffeine intake roughly the same; if you wish to stop taking caffeine, talk with your doctor about slowly weaning yourself from it.

How to take it

Take 200 mg of both caffeine and theanine (400 mg total) some 30 minutes before you need increased focus and attention.

People not used to caffeine should start with 50 mg of caffeine and 100 mg of theanine. Conversely, veteran coffee drinkers may need more than 200 mg of caffeine to experience its cognitive benefits, in which case the accompanying dose of theanine becomes 300 mg (even higher doses of theanine may not be dangerous, but neither have they shown greater benefits).

Supplementing caffeine on an empty stomach can increase the rate of absorption, but it can also cause stomach upset.

With its half-life of 5 to 6 hours, caffeine can disrupt sleep when consumed in the evening. Even if it doesn't prevent you from falling asleep, caffeine will impair the *quality* of your sleep.

Caffeine should be cycled based on its stimulatory effects. Should it no longer provide noticeable benefits, stop supplementation for 2 to 4 weeks to reset tolerance.

Secondary Options

Creatine

Why it's a secondary option

Creatine is an important source of fuel for cells, especially neurons. A genetic developmental disorder called "creatine transporter defect" (CTD) can cause a creatine deficiency and result in severe cognitive impairment. CTD can be treated with creatine supplementation.

More research is needed to determine whether creatine supplementation can benefit cognition in people who are not deficient. While most people produce enough creatine naturally to prevent cognitive complications, vegans and vegetarians might benefit from supplementation more than omnivores.

How to take it

Creatine monohydrate is both the most effective and least expensive form of creatine. Take 2–5 g with water. Active people should supplement closer to the high end of this range, though 2 g is technically sufficient to provide cognitive benefits.

If you are particularly sensitive to creatine's digestive side-effects, which include nausea and cramping, split your daily dose, take it with meals, and try *micronized* creatine monohydrate.

Cholinergics

Why they're a secondary option

Acetylcholine is a major neurotransmitter involved in memory formation. Elevated acetylcholine levels in the brain are associated with improved cognition. A supplement is said to be cholinergic when it increases the activity of acetylcholine, especially in the brain.

CDP-choline and alpha-GPC can provide the brain with the choline it needs to produce more acetylcholine. While choline bitartrate is much cheaper, little of it seems to reach the brain.

Huperzine-A can inhibit acetylcholinesterase, an enzyme that breaks down acetylcholine. Its half-life exceeds 24 hours, so it accumulates in the body when taken daily. It is problematic because of a dearth of long-term studies. There is a possibility that, with time, the body could adapt by producing more acetylcholinesterase, which would lead to reduced acetylcholine levels and a withdrawal period after huperzine-A supplementation has ceased. While the doses used in the studies (0.2–0.99 mg) were deemed safe in the short term, long-term supplementation cannot be recommended.

These supplements have been studied in the context of old-age dementia. Further research is needed to ascertain if they can improve cognition when used by younger people in good health.

How to take them

To supplement **alpha-GPC**, take 300–600 mg once a day with a meal. A higher dose (1,200 mg) can benefit people with dementia. Pairing alpha-GPC with uridine might provide synergistic benefits.

To supplement **CDP-choline**, take 250–500 mg twice a day (i.e. 500–1,000 mg/day).

To supplement **CDP-choline** for its pro-uridine content, take 500–1,000 mg twice a day (i.e. 1–2 g/day).

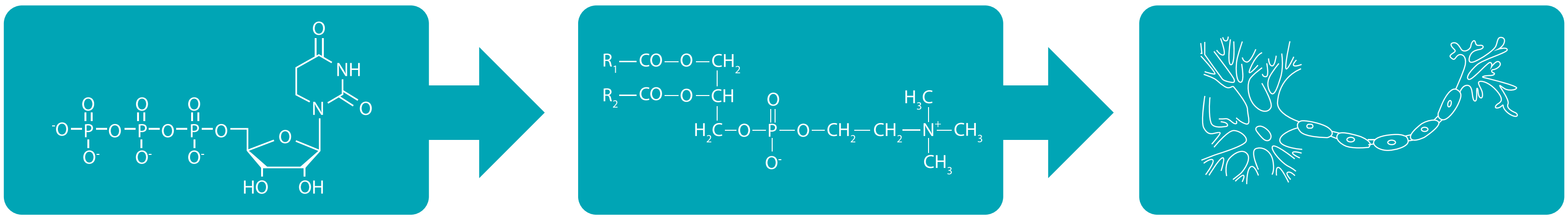
Uridine

Why it's a secondary option

Uridine is required to create neuronal membranes; it can also increase the rate of neuronal growth and turnover. There is little uridine in food. While the body can synthesize enough to satisfy its basic needs, preliminary evidence suggests that supplementation can benefit cognition.

Figure 3: How uridine could benefit cognition: mechanistic chart

Uridine | Phosphatidylcholine | Dendrite membranes



Uridine may be able to increase the brain levels of phosphatidylcholine.

Dendrites help neurons communicate with each other. Phosphatidylcholine is a component of dendritic membranes.

The incorporation of phosphatidylcholine into dendritic membranes is thought to benefit people suffering from a loss of synaptic function, as is seen in Alzheimer's disease.

Uridine has only been shown to be safe in the short term; more research is needed before long-term supplementation can be recommended. On the bright side, the benefits to cognition appear to be lasting (if new neurons are formed thanks to additional uridine, they do not just die off when supplementation is discontinued).

Uridine is thought to be synergistic with blueberry and *Bacopa monnieri*, though more research is needed to confirm this, too.

How to take it

Uridine is usually supplemented through uridine monophosphate (**UMP**), which is ⅔ uridine. The standard dosage for uridine is 250–500 mg twice a day (i.e. 500–1,000 mg/day), which translates as 375–750 mg of UMP twice a day (i.e. 750–1,500 mg/day).

Other options include triacetyluridine (TAU) and CDP-choline. Following oral supplementation, both appear to be effective in reaching the human brain.

If **TAU** is indeed seven times more bioavailable than an equimolar amount of uridine, as one study noted, then much lower doses could be used: 35–70 mg twice a day (i.e. 70–140 mg/day). Research on this compound is still scarce, however, which makes it hard to recommend.

As for **CDP-choline**, it is a source both of cytidine, which the body converts into uridine, and of choline, which the brain converts into acetylcholine. To supplement CDP-choline for its pro-uridine content, high doses are required: 500–1,000 mg twice a day (i.e. 1–2 g/day).

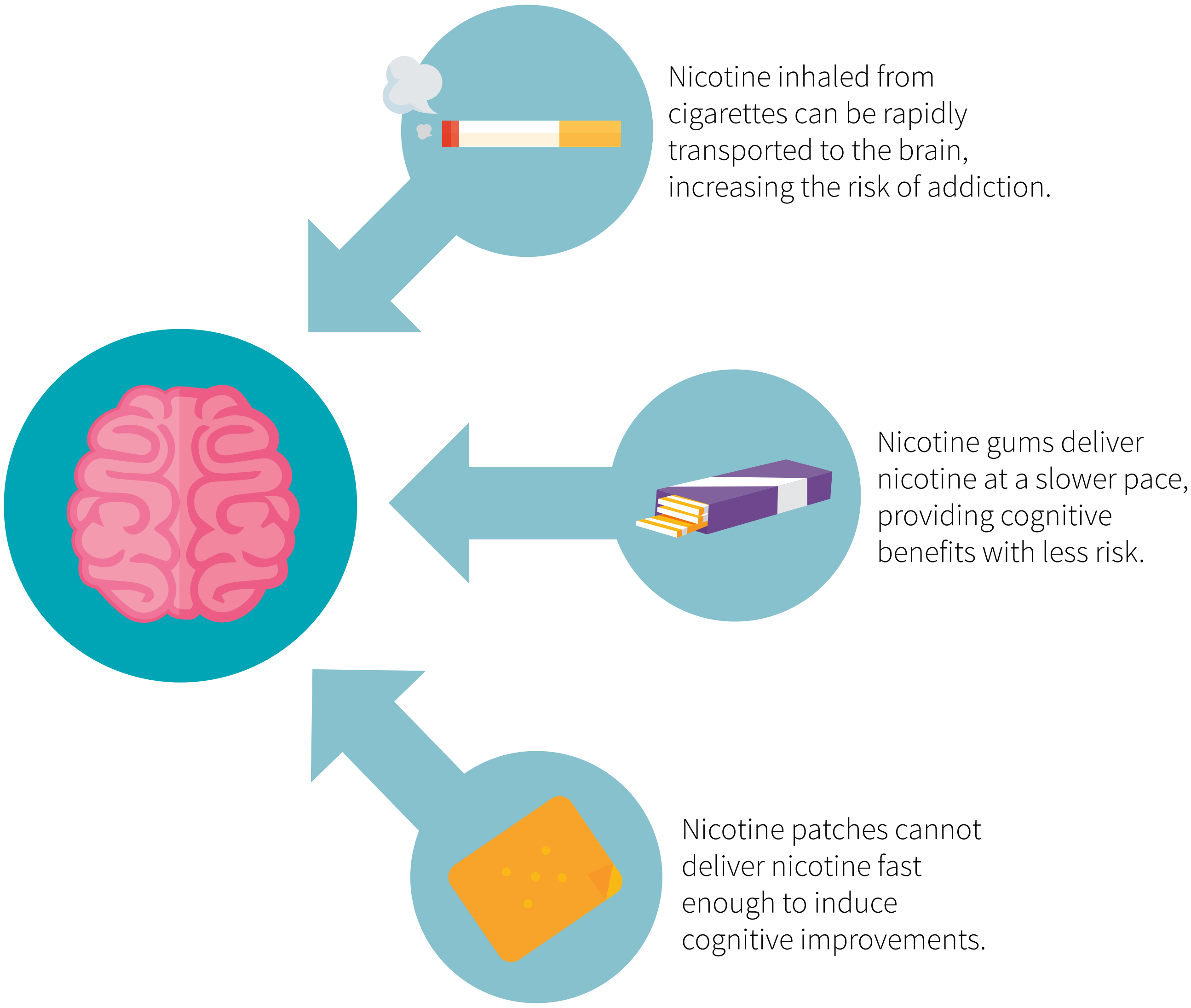
Further research is needed to determine if taking uridine with food is more effective than taking it on an empty stomach.

Inadvisable Supplements

Nicotine

Nicotine's addictive properties vary depending on the dose taken and the speed at which it enters the bloodstream. When inhaled, nicotine reaches the blood quickly, which makes this delivery method especially addictive. At the other end of the spectrum, patches are the least addictive delivery method, but they act too slowly to provide an acute stimulatory effect.

Figure 4: A model showing the benefits of nicotine on mental function



When it comes to speed of delivery, nicotine gum holds the middle ground. Chewing nicotine gum while working on a mentally demanding task (2 mg of nicotine at a time, no more than 10 mg in one day) has theoretical advantages. Making this a daily habit, however, would allow tolerance to develop, and only ceasing supplementation entirely (for a couple of weeks) would allow sensitivity to return. Increasing the dose instead would, sooner or later, lead to nicotine withdrawal. Even the minimum dose, taken regularly, is potentially addictive, and thus potentially harmful.

Of course, tobacco is still the most noxious source of nicotine, and not just because it contains some thirty carcinogens. As noted above, when inhaled, nicotine reaches the blood quickly, which makes it especially addictive. In addition, several other compounds in tobacco, such as monoamine oxidase inhibitors (MAOIs), amplify the addictive effects of nicotine. Finally, the acquired need to suck on something contributes to the addictive properties of cigarettes, cigars, and smoking pipes (and thumbs, for little children).

Coleus Forskohlii with Artichoke Extract

Long-term potentiation (LTP) refers to a persistent strengthening of synapses. Chemically induced long-term potentiation (CILTeP) refers to the initiation of LTP through the use of compounds that increase cell levels of cyclic adenosine monophosphate (cAMP). Elevated cAMP levels in brain cells are associated with improved cognition, memory formation, and muscle contractions.

The standard CILTeP stack combines forskolin (the bioactive in *Coleus forskohlii*), to increase cAMP production, with luteolin (the bioactive in artichoke extract), to inhibit cAMP deactivation. However, there is no evidence to suggest that this combination works. Forskolin and luteolin are flavonoids, which tend to be poorly absorbed. Most probably, neither compound reaches the brain to exert its desired effect.

Though forskolin and luteolin work in isolated brain cells, real-world supplementation does not increase cAMP levels. This "CILTeP stack" does not belong in any stack designed to improve memory and focus.

Assembling Your Stack

Incorporating Core Supplements

The only core supplement in the *Memory & Focus* stack is the humble ***blueberry***.

Choose one of the following protocols:

- Blueberry anthocyanins: 500–1,000 mg/day.
- Blueberry powder: 5.5–11 g/day.
- Fresh blueberries: 60–120 g/day.
- Blueberry juice: 500 mL/day.

Blueberries can be swapped for other dark berries. If you select the "juice" option, make sure your juice is made of actual berries, not of berry-flavored sugars!

The above protocols are recommended for most people; their efficacy and safety are backed by a significant body of evidence. Follow one of them for a couple of weeks before you consider making any modification, such as adding one of the following options.

Incorporating Options

For people who want to improve their long-term memory formation

Take the core blueberries, as described above. Take *Bacopa monnieri* (150 mg of bacosides) with a meal. Twice a day, take either:

- 500–1,000 mg of CDP-choline (i.e. 1–2 g/day).
- 150–300 mg of alpha-GPC (i.e. 300–600 mg/day) with 375–750 mg of UMP (i.e. 750–1,500 mg/day).
- 150–300 mg of alpha-GPC (i.e. 300–600 mg/day) with 35–70 mg of TAU (i.e. 70–140 mg/day).

For people who want to improve their attention span

Take the core blueberries, as described above. Take 200 mg of both caffeine and theanine (400 mg total) some 30 minutes before you need increased focus and attention.

People not used to caffeine should start with 50 mg of caffeine and 100 mg of theanine. Conversely, veteran coffee drinkers may need more than 200 mg of caffeine to experience its cognitive benefits, in which case the accompanying dose of theanine becomes 300 mg (even higher doses of theanine may not be dangerous, but neither have they shown greater benefits).

Caffeine should be cycled based on its stimulatory effects. Should it no longer provide noticeable benefits, stop supplementation for 2 to 4 weeks to reset tolerance.

Others options

People who don't frequently eat meat can add creatine (5 g taken with a meal) to any stack.

FAQ

Can I add to my stack a supplement not covered in this guide?

Supplement your current stack for a few weeks before attempting any change. Talk to your doctor and research each potential new addition in advance. Check for known negative interactions with other supplements in your current stack, but also for synergies. If two supplements are synergistic or additive in their effects, you might want to use lower doses for each.

Can I modify the recommended doses?

If a supplement has a recommended dosage range, stay within that range. If a supplement has a precise recommended dose, stay within 10% of that dose. Taking more than recommended could be counterproductive or even dangerous.

Do I take supplements with or without food? In the daytime or the evening?

Certain supplements have strong evidence for how and when to take them, which we have noted where applicable. For most supplements, however, this research is either mixed or absent. It is always a good idea to start with a low dose of a supplement, so as to minimize the harm of taking it during the day (e.g. tiredness) or in the evening (e.g. insomnia).

Keep in mind that taking a supplement with food does not mean that all foods will have the same effect on its absorption, digestion, or metabolism. The most basic example is fat-soluble vitamins (A, D, E, and K), which will absorb better with a small meal containing fat than a large meal containing little to no fat.

Can fish oil improve memory and focus?

Preliminary evidence suggests that fish oil may benefit cognition, but more research is needed before supplementation can be recommended for this specific purpose. Including more fatty fish in your diet may nevertheless be a good, healthy idea. Cod, salmon, and sardines are the best fish to eat, since they have high levels of omega-3 fatty acids but low levels of mercury. If you cannot change your diet and want to supplement fish oil, you can take a 1-g gelule daily.

Why are there no racetams in this stack?

While racetams (e.g. piracetam, aniracetam, oxiracetam) are thought to improve cognition, they are not recommended in this guide because they are pharmaceuticals and because human evidence is lacking. Further research is needed on their long-term safety and efficacy.

Precautions and Troubleshooting

Stack components are seldom studied together. The safest way to add supplements to your daily routine is one at a time, at least a couple of weeks apart, to better assess the effects (and side effects) of each new addition. Start at half the regular dose for a week, then slowly increase to the regular dose if you are not experiencing the desired effects.

Any supplement that can affect the brain, especially supplements with a stimulatory or sedative effect, should first be taken in a controlled situation. Do not take a dose, least of all your first dose, before events such as driving or operating heavy machinery, where impaired cognition may be a risk for your safety and the safety of others.